

Software engineer focused on high-performance web applications, real-time audio, and 2D/3D graphics. Built fintech products at Mobeewave, Apple, and Brainfinance, voice interfaces at ElevenLabs, and scientific visualization at Plotly. Current open-source work covers WebGL, WebAssembly, DSP, and color. Trained in control systems and signal processing.

CONTACT

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🌐 English, Russian, French (conversational), Spanish (basic)

WORK	Software Engineer Independent Oct 2025 – Present Open-source graphics, audio, and WebAssembly; selected projects total 3,000+ GitHub stars. - gl-waveform: WebGL2 time-series renderer; 60 fps over 172.8M samples per channel, exact per-pixel peaks, streaming input - audio: streaming audio engine; 10 GB+ files paged to OPFS, runs in a Worker; 300+ GitHub stars - JZ / watr: JavaScript-to-WebAssembly compiler and WAT optimizer, benchmarked against C, Rust, and Go - color-space: 168 verified color spaces for JS, WebAssembly, and shaders; 380+ GitHub stars - gpu-font: in-browser font identification on WebGPU; right family in the top five 92% of the time - Go backend (Gin, GORM, SQLite) for ISKCON Montréal Senior Frontend Engineer Brainfinance Nov 2022 – Oct 2025 Consumer fintech serving 100,000+ users in Canada: icash.ca loans and earna.zone savings. - Owned the earna.zone frontend on Next.js: landing, dashboard, application flows, and design system (shadcn, Tailwind, Storybook) - Rebuilt icash.ca loan application and contract signing; contract drop-off fell 40% in an A/B test - Raised key flows above 95 in PageSpeed; fixed WCAG AA accessibility issues Frontend Engineer (Contract) ElevenLabs Oct 2022 – Jan 2023 Concurrent pre-seed voice AI contract. - Launched elevenlabs.io: 11ty, Firebase auth, Stripe payments, voice API access - Built the browser player for generated speech Software Engineer Mobeewave / Apple Jan 2019 – Oct 2022 Joined Mobeewave in 2019 and continued at Apple after the 2020 acquisition. - Mobeewave: Samsung POS tap-to-pay frontend (PCI DSS), shipped in Samsung Wallet; acquirer portal used by 50+ financial institutions - Apple: Wallet web push provisioning, Apple Business Registry portal, Pay Later checkout widget - Reference polyfill for the W3C Template Instantiation proposal WebGL Engineer Plotly Sep 2016 – Sep 2018 Built and maintained the WebGL stack for plotly.js, an open-source scientific charting library. - Point clustering for real-time exploration of up to 1 billion points - WebGL scatter, line, scatterplot-matrix, and waveform renderers - Maintained regl (5,500+ GitHub stars) and 24 gl-vis repositories - Visual-regression tooling for 2,000+ chart variants
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OPEN SOURCE	AudioJS 70+ audio and DSP repositories for browsers and Node. - web-audio-api: Web Audio API for Node; 950+ GitHub stars - digital-filter: IIR, FIR, adaptive, and multirate filters Wavefont / Linefont Fonts that draw waveforms and line charts as text; 900+ GitHub stars combined.
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sprae / Spect
sprae: reactive HTML attributes; made earna.zone server components interactive without client components. Spect: state and component framework.

EARLIER EXPERIENCE	Web Developer TTBA Group 10+ sites at 100/100 PageSpeed; Wishbox.gift on Preact.	Jan 2015 – Aug 2016
	Frontend Developer KudaGo City guide for 15+ cities; mobile site under 20 KB of JavaScript, 100/100 PageSpeed at launch.	Mar 2012 – Jan 2015

EDUCATION	Baltic State Technical University "Voenmeh" BSc + MSc BSc: Information & Computer Science; MSc: Computer Graphics & Intelligent Systems	Sep 2006 – Jun 2012
	Saint Petersburg College of Marine Instrumentation Technician diploma Automatic Control Systems · Signal Processing · Control Theory · Electronics · Instrumentation	Sep 2002 – Jun 2006

SKILLS	Graphics & WebAssembly WebGL WebGPU GLSL regl Canvas SVG three.js D3 WebAssembly SIMD	Audio & DSP Web Audio API neural audio voice synthesis DSP FFT IIR/FIR filters codecs streaming LUFS
	Frontend TypeScript React Next.js Preact Vue signals web components SSR	Design Systems & UI shadcn Tailwind Storybook Figma responsive UI WCAG / ARIA A/B testing
	Performance & Quality profiling Web Workers Core Web Vitals Lighthouse visual regression Playwright Cypress Puppeteer Jest	Platform & Delivery Node.js Go Python GraphQL Firebase PostgreSQL CI/CD GitHub Actions esbuild / Rollup